Diamond clause of the canonical model: possibility witnessed by an accessible maximal world

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-normal-modal-logic-completeness/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: claim_key carries @1256: the identical label recurs at line 1321 in the both-primitive branch; that instance is a separate card (L46141).

Editorial note. The label prop:diamond names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Editorial note. OLP reuses this label inside the unit; the card is identified by its statement span (source lines 1256-1261) and the claim key carries the statement start line.

Allowed standard prerequisites (editorial).

- Box2 transfer lemma. Same section, \ollabel{lem:box2} (source line 1170, own card). Line 1272: non-derivability of Box not-A at Gamma becomes non-derivability of not-A at the inverse image.
- Consistency existence fact (adding a formula). Proof theory part, prop:consistencyfacts item (b). Line 1274: the inverse image plus A is Sigma-consistent.
- Lindenbaum's lemma. Same part, \ollabel{thm:lindenbaum} (own card). Line 1275: completion to the witnessing world Delta.

Source reference. Original lemma lem-box2: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 1214–1214. Prerequisite pointer invoked at proof line(s) 1276; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-consistencyfacts: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 1278–1278. Named citation invoked at source line(s) 1278: the author's marker \olref[prf][con]{prop:consistencyfacts} resolves inside the full Open Logic Project to another unit (not staged in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-consistencyfacts-b: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 1278–1278. Named citation invoked at source line(s) 1278: the author's marker \olref[prf][con]{prop:consistencyfacts-b} resolves inside the full Open Logic Project to another unit (not staged in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *If Γ is complete Σ-consistent, then $\Diamond A \in \Gamma$ if and only if for some complete Σ-consistent Δ such that $\Diamond\Delta \subseteq \Gamma$, it holds that $A \in \Delta$.*

Proof. Suppose Γ is complete Σ-consistent. The right-to-left part is easy: Let Δ be complete Σ-consistent such that $\Diamond\Delta \subseteq \Gamma$ and $A \in \Delta$. Then $\Diamond A \in \Diamond\Delta \subseteq \Gamma$, i.e., $\Diamond A \in \Gamma$.

For the left-to-right direction, assume $\Diamond A \in \Gamma$. We have to show that there is a complete Σ-consistent Δ with $A \in \Delta$ and $\Diamond\Delta \subseteq \Gamma$. Since $\Diamond A \in \Gamma$, and Γ is Σ-consistent, $\Box\neg A \notin \Gamma$. Since Γ is complete Σ-consistent, it is deductively closed, so $\Gamma \vdash /[\Sigma]\Box\neg A$. By lem-box2, $\Box^{-1}\Gamma \vdash /[\Sigma]\neg A$. By prop-consistencyfactsprop-consistencyfacts-b, $\Box^{-1}\Gamma \cup \{A\}$ is Σ-consistent. By Lindenbaum's Lemma, there is a complete Σ-consistent set Δ such that $\Box^{-1}\Gamma \cup \{A\} \subseteq \Delta$. Clearly, $A \in \Delta$. To see that $\Diamond\Delta \subseteq \Gamma$, assume it weren't: For some $B \in \Delta$, $\Diamond B \notin \Gamma$. Then, since Γ is complete Σ-consistent, $\Box\neg B \in \Gamma$. But then also $\neg B \in \Box^{-1}\Gamma \subseteq \Delta$, which would make Δ inconsistent. $\Box$